

Rothschild on anaphora

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1. Two kinds of donkey sentences

Anaphora from scope to restrictor: pronoun (or definite description) in nuclear scope of quantifier anaphoric on an indefinite in restrictor

- (1) Every farmer who owns a donkey beats it / the donkey / that donkey

Anaphora across a conjunction: pronoun or DD in a second conjunct anaphoric on an indefinite in the first conjunct. Conjunction can be in nuclear scope:

- (2) Every farmer owns a donkey and beats it / the donkey / that donkey
(3) Every man met a woman and liked her

or in restrictor:

- (4) Every farmer who owns a donkey and beats it is a sadist

A moral of both papers under discussion is that anaphora across conjunction is really important.

- ‘Projection from Situations’ [PFS] points out that one of the leading theories of this topic (the situation E-type theory) totally flounders with anaphora across conjunction; points out what would need to be done to solve this problem; argues that it’s not very attractive; and also makes some persuasive independent objections to the program.
- ‘A Trivalent Approach to Anaphora and Presupposition’ [ATAAP] develops a theory, drawing on some ideas associated with the dynamic tradition (but not the idea that semantic values of sentences are functions or relations on some set), which in the first instance only has an account of anaphora across conjunction, but ends up by proposing a bold maneuver that allows assimilating other kinds of anaphora (including scope-to-restrictor anaphora and anaphora across disjunction) to anaphora across conjunction.

1. Some initial contrasts

- Cross-conjunction examples clearly lack uniqueness presuppositions—e.g. (3) leaves it totally open that some men met more than one woman. With the scope-to-restrictor examples, that’s the assumption in the literature, but I have aberrant reactions to some cases:

- (5) a. Every man met 1 or 2 women. Every man met a woman and liked her.

- b. ? Every man met 0, 1, or 2 women. Every man who met a woman liked her.

Exception: when the pronoun is replaced by a rich description like ‘the donkey he owns’, I get strong uniqueness presuppositions in both kinds of examples. (This is *prima facie* problematic for Heim 1990’s view on which the pronouns are equivalent to these kinds of rich descriptions.)

- With scope-to-restrictor there is the contrast between weak (every farmer who owns a donkey beats some donkey he owns) versus strong (every farmer who owns a donkey beats every donkey he owns) truth conditions, and a haziness about the truth conditions when the quantifier is something like ‘most’. With cross-conjunction anaphora there is no parallel issue: ‘met a woman and liked her’ seems interchangeable with ‘met and liked a woman’.
 - Across a conjunction the indefinite has to precede the thing that’s anaphoric on it. With type A it’s less clear that we can’t go the other way round.
- (6) a. * Every farmer owns it and beats a donkey
 b. ?? Every farmer who owns it beats a donkey
 b. ? Every pilot who shot at it hit a MiG

2. Rothschild and Mandelkern on the Situation E-type theory

The general idea of an ‘E-type’ or ‘D-type’ theory: pronouns equivalent to definite descriptions understood Frege-style or Russell-style.

Scope-to-restrictor anaphora poses two challenges for this approach: first, it seems to predict an unwanted uniqueness presupposition for (1); second, even if we just allowed this to be cancelled (treating the definite description like an existential quantifier) we would still only get the weak truth condition.

Situation E-type theorists have developed fancy theories of quantifiers like ‘every’ designed to address both problems.

Loosely: (1) means that for every object *o* and minimal situation *m* in which *o* is a farmer who owns a donkey, *o* beats the donkey *in m*.

Büding: backreference to *m* achieved by a special variable *r* on ‘the donkey’ which the clause for ‘every’ directly manipulates.

Until Rothschild and Mandelkern came along to offer them a similarly fancy theory of ‘and’, there was no version of the situation E-type theory that could avoid unwanted uniqueness presuppositions for cross-conjunction donkey sentences like (2–4).

- Worse still: in views like Elbourne’s and Büding’s where ‘it’ means ‘the donkey’ rather than something longer like ‘the donkey he owns’, we either get a ridiculously

strong truth condition for (2) where all the farmers own the same donkey, or a ridiculously weak reading where they only have to beat some donkey, not necessarily any donkey they own!

Rothschild-Mandelkern situational account of ‘John met a woman and liked the woman’: there is a minimal situation m in which John met a woman, and there is a situation in which John liked the woman in m .

More generally: a conjunction is true on an assignment f if there’s a minimal situation m on which the first conjunct is true, and the second conjunct is true on $f[r \mapsto m]$. (I’m making explicit the element of existential quantification which M&R leave implicit.)

$$\llbracket \phi \wedge \psi \rrbracket^{\mathcal{L}^s} = 1 \text{ iff for some minimal } m \leq s, \llbracket \phi \rrbracket^{\mathcal{L}^m} = 1 \text{ and } \llbracket \psi \rrbracket^{f[r \mapsto m], s} = 1$$

How cheesy and *ad hoc* is it to think of ‘and’ as a variable binder like this? Pretty bad...

- Note however that the problem about minimality M&R raise in section 8.2 naturally motivates the shift from an ‘inexact’ situation semantics to a Fine-style ‘exact’ situation semantics. In the latter, the clause for conjunction is *already* bringing in an element of existential quantification: s exactly verifies $\phi \wedge \psi$ on f iff for some s' and s'' , $s = s' + s''$ and s' exactly verifies ϕ on f and s'' exactly verifies ψ on f . In this setting we can provide for anaphora just by replacing the final ‘ f ’ with ‘ $f[r \mapsto s']$ ’.

3. Another contrast between cross-conjunction and scope-to-restrictor: the viability of an old-school scope-based approach

A different way for a broadly E-type-like theory to handle sentences like (2–4) treats indefinites like ‘a donkey’ are treated simply as existential quantifiers which are allowed to take wide scope in environments where other quantifiers can’t.

$$(2') \quad [\text{Every farmer}] \lambda x_1. [[\text{a donkey}] \lambda x_2. [x_1 \text{ owns } x_2 \text{ and beats it}_2]]$$

Scope-to-restrictor donkey sentences can’t be handled like this.

$$(1') \quad [\text{A donkey}] \lambda x_1. [[\text{every farmer who owns } x_1] \lambda x_2. [x_2 \text{ beats it}_1]]$$

(1') has a truth condition (there’s a donkey all of whose owners beat it) that may be a possible reading for (1) but is certainly not the main one.

The idea that indefinites can escape scope islands that trap other quantifiers like ‘every F ’ and ‘most F s’ is independently motivated:

- (7) a. If an uncle of mine dies I will inherit a fortune (Fodor and Sag)
- b. Every professor rewarded every student who read some book (Abusch)

Some (Schulz ***) advocate simply embracing these appearances. Others have crafty ways of getting the effects of wide scope using different mechanisms.

Note that simply going for a Heim / Kamp / Rothschild-style view on which indefinites work like variables and the existential-quantification-like effects are generated elsewhere does not by itself take care of this phenomenon. For example with (7b) these techniques can get a reading like this:

(7b') [Every professor] $\lambda x_1. [\exists x_2 [[\text{every } \lambda x_3. [\text{student}(x_3) \wedge \text{book}(x_2) \wedge x_3 \text{ read } x_2]] [x_1 \text{ rewarded } x_3]]]$

This isn't what we want, since the existence of any non-book is sufficient for it to be true (or at least non-false) (Kratzer, 'Scope or Pseudoscope').

So, apparently wide-scoping indefinites are a hard challenge for everyone; it's unclear whether the distinctive innovations of the dynamic tradition really help with the hard challenge as a whole, although they do help with the easy special case of getting 'met a woman and liked her' to be semantically equivalent to 'met and liked a woman'.

4. A strategy for extending theories of cross-conjunction anaphora to other kinds

At a certain point towards the end of ATAAP, Daniel has constructed a semantic system that handles cross-conjunction anaphora fine (i.e. it makes 'he met a woman and he liked her' interchangeable with 'he met and liked a woman'), but does not deal with other cases of anaphora, including scope-to-restrictor anaphora and anaphora across disjunction as in Partee's

(8) Either this house doesn't have a bathroom or it's under the stairs

To extend the reach of the account to encompass these cases, Daniel makes a bold move: he allows a kind of transformation rule in which, en route to the structure that is the input to compositional semantics, we can replace sentences with "classical equivalents" (to which we will then go on to apply a further rule of '†-insertion' analogous to Heim's 'existential closure'). In particular, we can make the following replacements:

$$\phi \vee \psi \rightarrow \phi \vee ((\neg\phi) \wedge \psi)$$

$$\text{Every}_x(\phi, \psi) \rightarrow \text{Every}_x(\phi, \phi \wedge \psi)$$

$$\text{If } \phi, \psi \rightarrow \text{If } \phi, \phi \wedge \psi$$

The move lets us replace (1) with 'Every farmer who owns a donkey owns a donkey and beats it'. It thus gets us the 'weak' truth condition for (1); something else needs to be added to the story to get the 'strong' reading.

- This is worthy of careful exploration even if you find it methodologically suspicious, since it's possible the effect of this could be got without actual syntactic manipulations.
- Note that that insofar as the rewriting move is available, it's also available to those who have other techniques for getting 'met a woman and liked her' to mean the

same as ‘met and liked a woman’, including interpreting ‘a woman’ as a boring wide scope existential-quantifier.

5. Some details on the semantic system in ATAAP

Assignment functions: total but can map variables to the ‘impossible object’ $\perp$.

Semantic values of sentences: functions from $\langle \text{world}, \text{assignment} \rangle$ pairs to $\{1, \frac{1}{2}, 0\}$.

Semantic value of ‘liked(x,y)’ (corresponding to ‘he_x liked her_y’): maps $\langle w, a \rangle$ to $\frac{1}{2}$ iff $a(x) = \perp$ or $a(y) = \perp$. Otherwise, maps $\langle w, a \rangle$ to 1 iff $a(x)$ liked $a(y)$ at w .

Semantic value of ‘some_y(woman(y), met(x,y))’ (corresponding to ‘he_x met a woman_y’): maps $\langle w, a \rangle$ to $\frac{1}{2}$ iff $a(x) = \perp$. Maps $\langle w, a \rangle$ to 0 whenever $a(y) = \perp$. Otherwise, maps $\langle w, a \rangle$ to 1 iff $a(x)$ met $a(y)$ at w .

Two possibilities for ‘and’, Strong and Middle Kleene.

- Strong Kleene: $\llbracket \phi \wedge \psi \rrbracket^{w,a} = \min(\llbracket \phi \rrbracket^{w,a}, \llbracket \psi \rrbracket^{w,a})$
- Middle Kleene: $\frac{1}{2}$ if $\llbracket \phi \rrbracket^{w,a} = \frac{1}{2}$, $\min(\llbracket \phi \rrbracket^{w,a}, \llbracket \psi \rrbracket^{w,a})$ otherwise

So: $\llbracket \text{some}_y(\text{woman}(y), \text{met}(x,y)) \wedge \text{liked}(x,y) \rrbracket^{w,a} =$

$\frac{1}{2}$ if $a(x) = \perp$

1 if $a(y)$ is a woman at w and $a(x)$ met $a(y)$ at w and $a(x)$ liked $a(y)$ at w

0 otherwise (including the case where $a(y) = \perp$)

$\llbracket \text{some}_y(\text{woman}(y), \text{met}(x,y) \wedge \text{liked}(x,y)) \rrbracket$ is the same.

For donkey sentences like (2) we need a freely-addable existential closure operator ‘ $\dagger$ ’:

(2) is represented as

(2'') Every_x (farmer(x), $\dagger \text{some}_y(\text{owns}(x,y), \text{beats}(x,y))$)

$\llbracket \dagger \phi \rrbracket^{w,a} =$

$\max\{\llbracket \phi \rrbracket^{w,a'} : a' \text{ differs from } a \text{ only on variables that are **definedness-insensitive** in } \phi\}$

where

v is **definedness-insensitive** in $\phi =_{\text{df}}$ whenever $\llbracket \phi \rrbracket^{w,f[v \mapsto \perp]} = \frac{1}{2}$, $\llbracket \phi \rrbracket^{w,f[v \mapsto x]} = \frac{1}{2}$ for all x .

Upshot: variables associated with *indefinites* (subscripts to ‘some’) in ϕ get existentially bound in $\dagger \phi$. Variables associated only with pronouns (and other definites) do not.¹

Note that this limitation on the freely added existential quantifier is crucial, otherwise ‘Every farmer beats it’ could mean ‘Every farmer beats something’!

- Q: is the fact that $\dagger$ indiscriminately binds *all* the definedness-insensitive variables in its scope going to be a problem? Maybe yes, if we want a reading of ‘Every farmer

¹ Except in weird cases like $x=x \wedge \text{John knows } 1+1=3$, where it doesn’t seem to matter.

filled in a form he was given by a (universally-loathed) taxman' to have a reading where there has to be a single taxman but every farmer can have a different form.

6. Dynamic “post-semantics”

One *could* combine the above semantics with a standard picture where open sentences (i.e. ones that have different truth values on different assignments relative to some world) are either unassertible, or can be asserted only when there's a contextually salient assignment function for the free variables.

Instead, Daniel has a “dynamic” picture of the communicative role of open sentences.

Contexts: sets of $\langle \text{world}, \text{assignment} \rangle$ pairs.

Felicity rule: ϕ is felicitous in c iff $\llbracket \phi \rrbracket^{f,w} \neq \frac{1}{2}$ for every $\langle f, w \rangle \in c$

Update rule: when sentence ϕ is asserted (and accepted) in context c the new context is $\{ \langle w, a \rangle \in c : \llbracket \phi \rrbracket^{w,a} = 1 \}$.

Starting context: contains every $\langle w, a \rangle$ pair (where the world conforms to background knowledge).

- One could probably achieve equivalent effects by having the starting context contain only the “empty” assignment (mapping everything to $\perp$), as is traditional, but this would certainly look uglier.
- Note how this .

Motivation: allow anaphora to escape the boundaries of the sentence, as in:

(9) A man_x came in. He_x sat down.

‘He_x sat down’ is infelicitous in the starting context, but felicitous after ‘A man_x came in’ knocks out all every $\langle w, a \rangle$ where $a(x) = \perp$ (or $a(x)$ isn't a man who came in a t w).

Some qualms I feel about this framework:

- It's quite unclear how to think about the speech act performed when you knock some $\langle w, a \rangle$ pairs out of the context set as a matter of *saying* that something is the case. But surely we do typically say things when we utter declarative sentences.
- Knocking out worlds requires really changing minds; knocking out assignments is a totally different kind of activity with no psychological reality outside the current conversation; is it wise to lump them together?
- Don't anaphoric follow ups assume that the speaker ‘had some particular thing in mind’, in a way that's left out by this picture? (‘I saw a sloth yesterday — Was it a three-toed sloth?’)

- obvious advantage of the ‘text’ approach: there are lots of other things that work like this. E.g. multi-sentence conditionals: ‘If he’s right, peak oil is near. There will soon be an economic collapse and we’d better buy canned goods. But his track record doesn’t suggest much reason to worry’. Or multi-sentence universal quantification over times: ‘Every time I make soufflé there’s some kind of disaster. The soufflé doesn’t rise or takes forever to cook’.
- Something the ‘text’ approach doesn’t give you: pronouns with a range of “candidate referents” that can be successively narrowed down by different conversational participants. But is this a language game we actually play?

A: I saw a sloth yesterday.

B: Was it a three-toed sloth?

A: No, it was a two-toed sloth, but I did also see a three-toed sloth yesterday.